

Object Oriented Programming, Group C Assignment No.9

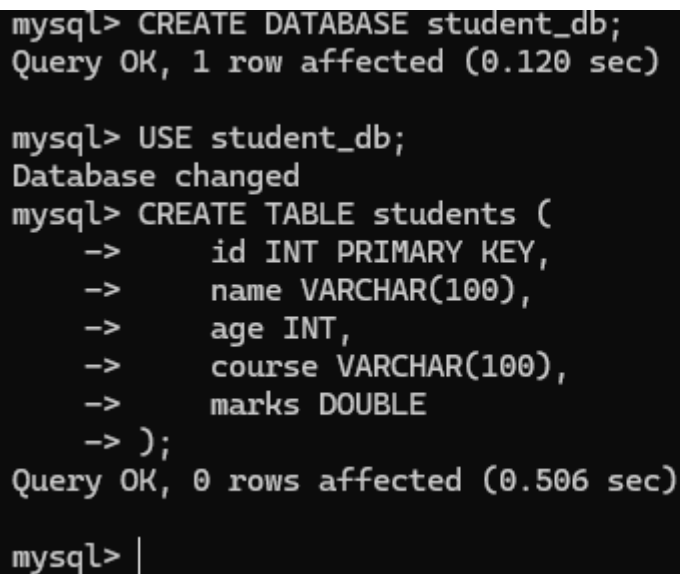
Steps:

1.Create a Database student_db in MySQL using following queries

```
CREATE DATABASE student_db;  
USE student_db;
```

2.Create Tables In Database

```
CREATE TABLE students (  
    id INT PRIMARY KEY,  
    name VARCHAR(100),  
    age INT,  
    course VARCHAR(100),  
    marks DOUBLE  
);
```



```
mysql> CREATE DATABASE student_db;  
Query OK, 1 row affected (0.120 sec)  
  
mysql> USE student_db;  
Database changed  
mysql> CREATE TABLE students (  
    -> id INT PRIMARY KEY,  
    -> name VARCHAR(100),  
    -> age INT,  
    -> course VARCHAR(100),  
    -> marks DOUBLE  
    -> );  
Query OK, 0 rows affected (0.506 sec)  
  
mysql> |
```

3.Download JDBC Driver

<https://dev.mysql.com/downloads/connector/j/>

MySQL → mysql-connector-j

4.Write a following JAVA code

```

import java.sql.*;
import java.util.Scanner;

public class StudentManagement {

    static final String URL = "jdbc:mysql://localhost:3306/student_db";
    static final String USER = "root";
    static final String PASSWORD = "password"; // change

    static Connection con;

    // CONNECT DATABASE
    public static void connectDB() {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            con = DriverManager.getConnection(URL, USER, PASSWORD);
            System.out.println("Connected to DB!");
        } catch (Exception e) {
            System.out.println("Connection Error: " + e.getMessage());
        }
    }

    // CREATE (ADD STUDENT) with TRANSACTION
    public static void addStudent(int id, String name, int age, String course, double marks) {
        String sql = "INSERT INTO students VALUES (?, ?, ?, ?, ?)";

        try {
            con.setAutoCommit(false); // START TRANSACTION

            PreparedStatement ps = con.prepareStatement(sql);
            ps.setInt(1, id);

```

```

        ps.setString(2, name);
        ps.setInt(3, age);
        ps.setString(4, course);
        ps.setDouble(5, marks);

        ps.executeUpdate();

        con.commit(); // COMMIT

        System.out.println("Student Added!");

    } catch (Exception e) {
        try {
            con.rollback(); // ROLLBACK if error
        } catch (Exception ex) {
            System.out.println("Rollback Error: " + ex.getMessage());
        }
        System.out.println("Error: " + e.getMessage());
    } finally {
        try {
            con.setAutoCommit(true);
        } catch (Exception e) {}
    }
}

// READ ALL

public static void viewStudents() {
    try {
        String sql = "SELECT * FROM students";
        Statement st = con.createStatement();
        ResultSet rs = st.executeQuery(sql);

        System.out.println("\n--- Student List ---");
    }
}

```

```

while (rs.next()) {
    System.out.println(
        rs.getInt("id") + " | " +
        rs.getString("name") + " | " +
        rs.getInt("age") + " | " +
        rs.getString("course") + " | " +
        rs.getDouble("marks")
    );
}

} catch (Exception e) {
    System.out.println(e);
}
}

// READ ONE
public static void getStudent(int id) {
    try {
        String sql = "SELECT * FROM students WHERE id=?";
        PreparedStatement ps = con.prepareStatement(sql);
        ps.setInt(1, id);

        ResultSet rs = ps.executeQuery();

        if (rs.next()) {
            System.out.println("\n--- Student Info ---");
            System.out.println("ID: " + rs.getInt("id"));
            System.out.println("Name: " + rs.getString("name"));
            System.out.println("Age: " + rs.getInt("age"));
        }
    }
}

```



```

    } catch (Exception e) {
        try { con.rollback(); } catch (Exception ex) {}
        System.out.println("Error: " + e.getMessage());
    } finally {
        try { con.setAutoCommit(true); } catch (Exception e) {}
    }
}

```

// DELETE with TRANSACTION

```

public static void deleteStudent(int id) {
    String sql = "DELETE FROM students WHERE id=?";

    try {
        con.setAutoCommit(false);

        PreparedStatement ps = con.prepareStatement(sql);
        ps.setInt(1, id);

        int rows = ps.executeUpdate();

        if (rows > 0) {
            con.commit();
            System.out.println("Deleted successfully!");
        } else {
            con.rollback();
            System.out.println("Student not found!");
        }
    }

    } catch (Exception e) {
        try { con.rollback(); } catch (Exception ex) {}
        System.out.println("Error: " + e.getMessage());
    }
}

```

```
    } finally {  
        try { con.setAutoCommit(true); } catch (Exception e) {}  
    }  
}
```

```
// MAIN MENU
```

```
public static void main(String[] args) {
```

```
    connectDB();
```

```
    Scanner sc = new Scanner(System.in);
```

```
    int choice;
```

```
    do {
```

```
        System.out.println("\n===== STUDENT MANAGEMENT =====");
```

```
        System.out.println("1. Add Student");
```

```
        System.out.println("2. View All Students");
```

```
        System.out.println("3. Get Student");
```

```
        System.out.println("4. Update Student");
```

```
        System.out.println("5. Delete Student");
```

```
        System.out.println("6. Exit");
```

```
        choice = sc.nextInt();
```

```
        switch (choice) {
```

```
            case 1:
```

```
                System.out.print("ID: ");
```

```
                int id = sc.nextInt();
```

```
                sc.nextLine();
```

```
                System.out.print("Name: ");
```

```
                String name = sc.nextLine();
```

```
System.out.print("Age: ");
```

```
int age = sc.nextInt();
```

```
sc.nextLine();
```

```
System.out.print("Course: ");
```

```
String course = sc.nextLine();
```

```
System.out.print("Marks: ");
```

```
double marks = sc.nextDouble();
```

```
addStudent(id, name, age, course, marks);
```

```
break;
```

```
case 2:
```

```
viewStudents();
```

```
break;
```

```
case 3:
```

```
System.out.print("Enter ID: ");
```

```
getStudent(sc.nextInt());
```

```
break;
```

```
case 4:
```

```
System.out.print("Enter ID: ");
```

```
id = sc.nextInt();
```

```
sc.nextLine();
```

```
System.out.print("Name: ");
```

```
name = sc.nextLine();
```

```
System.out.print("Age: ");
```

```
age = sc.nextInt();  
sc.nextLine();
```

```
System.out.print("Course: ");  
course = sc.nextLine();
```

```
System.out.print("Marks: ");  
marks = sc.nextDouble();
```

```
updateStudent(id, name, age, course, marks);  
break;
```

case 5:

```
System.out.print("Enter ID: ");  
deleteStudent(sc.nextInt());  
break;
```



```
PS C:\Users\Software Lab -01\Desktop\11.Java\std> & 'C:\Program Files\Java\jdk-17\bin\java.exe' '@C:\Users\Software Lab -01\AppData\Local\Temp\cp_91t7hmujuztz6d4loumKSt9_argfile' 'StudentManagement'
Connected to DB!

===== STUDENT MANAGEMENT =====
1. Add Student
2. View All Students
3. Get Student
4. Update Student
5. Delete Student
6. Exit
1
ID: 001
Name: Harry
Age: 20
Course: AIDS
Marks: 95
Student Added!

===== STUDENT MANAGEMENT =====
1. Add Student
2. View All Students
3. Get Student
4. Update Student
5. Delete Student
6. Exit
2

--- Student List ---
1 | Harry | 20 | AIDS | 95.0

===== STUDENT MANAGEMENT =====
1. Add Student
2. View All Students
3. Get Student
4. Update Student
5. Delete Student
6. Exit
3
Enter ID: 1

--- Student Info ---
ID: 1
Name: Harry
Age: 20
Course: AIDS
Marks: 95.0

===== STUDENT MANAGEMENT =====
1. Add Student
2. View All Students
3. Get Student
4. Update Student
5. Delete Student
6. Exit
```

8. In MySQL check for the insert data using following queries

SHOW DATABASES;

USE student_db;

SHOW TABLES;

SELECT * FROM students;

```
mysql> SELECT * FROM students;
+----+-----+-----+-----+-----+
| id | name  | age  | course | marks |
+----+-----+-----+-----+-----+
| 1  | Harry | 20   | AIDS   | 95    |
+----+-----+-----+-----+-----+
1 row in set (0.009 sec)
```